A-A-51166
April 30, 1987

COMMERCIAL ITEM DESCRIPTION

FILES, ROTARY, INVERTED CONE SHAPE, HIGH SPEED STEEL OR TUNGSTEN CARBIDE

The General Services Administration has authorized the use of this commerical item description in preference to the type VIII rotary file covered by interim Federal Specification GGG-F-00340.

1. SCOPE

1.1 Scope. This description covers inverted cone shape high speed steel or tungsten carbide rotary files of the standard heavy duty commercial type commonly used in an industrial environment. Figure 1 depicts the shape of these files.

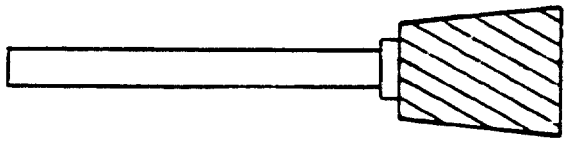


Figure 1. Illustration of inverted cone shape rotary file.

1.2 Classification. Rotary files shall be of the following types, styles, sizes, and classes. The type, style, size, and class to be furnished for a specific procurement shall be as specified (see 6.1).

Type I - High speed steel.
Type II - Tungsten carbide.

Style A - Right hand spiral fluting.
Style B - Right hand spiral fluting with chip breaker.
Style C - Double cut fluting.
Style D - Diamond cut fluting.

Size - Sizes shall be selected from tables I and II for the type rotary file specified.

Class a - Coarse tooth spacing (see tables III and IV).
Class b - Medium tooth spacing (see tables III and IV).
Class c - Fine tooth spacing (see tables III and IV).

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TABLE I. Type I, high speed steel rotary files sizes and characteristics.

Size	Characteristics		
	Shank diameter (inches)	Cutter head dimensions (inches)	
		diameter	length
1	1/8	1/4	1/4
2	1/4	1/4	1/2
3	1/4	1/2	1/2
4	1/4	1/2	3/4
5	1/4	3/4	5/8

TABLE II. Type II, tungsten carbide rotary files sizes and characteristics.

Size	Characteristics		
	Shank diameter (inches)	Cutter head dimensions (inches)	
		diameter	length
1	1/8	1/8	3/16
2	1/8	3/32	1/8
3	1/8	1/8	1/8
4	1/8	1/4	1/4
5	1/4	1/4	5/16
6	1/4	3/8	3/8
7	1/4	1/2	1/2
8	1/4	3/4	5/8

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this description to the extent specified herein.

Federal Specification PPP-T-1150 - Tools and Tool Accessories for Power Driven, Metal and Wood-working Machinery, Packaging and Packing of.

Federal Standard FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

American Society for Testing and Materials ASTM A600 - Standard Specification for High Speed Tool Steel.

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American Society for Testing and Materials ASTM D 3951 - Standard Practice for Commercial Packaging.

3. SALIENT CHARACTERISTICS

3.1 Design. The rotary files shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The rotary files shall meet the requirements specified herein and shall conform to current industrial practice.

3.2 Materials and hardness. The cutting heads of type I rotary files shall be manufactured from ASTM A600 high speed tool steel with a hardness of not less than 62 and not greater than 65 on the Rockwell "C" scale. The cutting heads of type II rotary files shall be manufactured from C-2 commercial grade tungsten carbide, with a hardness of not less than 91.5 and not greater than 92.5 on the Rockwell "A" scale. Type I and II rotary files shall either be a one or two piece construction. Where one piece construction is used, type I rotary files shall be of solid, high speed steel construction and type II rotary files shall be of solid tungsten carbide construction. Where a two piece construction is used, types I and II rotary files shall have a steel shank securely joined to the cutting head in a manner which conforms to the manufacturer's standard commercial practice. The shanks of type I rotary files shall have a hardness conforming to the manufacturer's standard part drawing for the rotary file specified in the contract or purchase order. The shanks of type II rotary files that are of a two piece construction shall be hardened and tempered to a hardness of not less than 30 and not greater than 55 on the Rockwell "C" scale.

3.3 Head and shank concentricity. The head and shank shall be concentric to within 0.005 inch total indicator reading (TIR).

3.4 Head and shank tolerances. The allowable shank diameter tolerance shall be +0.000 inch, -0.002 inch. The shank length shall conform to the manufacturer's standard part drawing for the rotary file specified in the contract or order, provided that 1/8 inch diameter shanks shall be not less than 1 inch in length, and 1/4 inch diameter shanks shall be not less than 1-1/2 inches in length. The allowable tolerance for cutter head diameters shall be not greater than ±0.010 inch. The allowable cutter head length tolerance shall be not greater than ±0.062 inch.

3.5 Number of teeth per circumference. The number of cutting teeth on the circumference of the rotary file shall conform to table III or table IV for the type, diameter, and class rotary file required. The tooth count per circumference shall be the number of right hand spirals on the circumference of the rotary file. Where ranges are shown in table III and table IV for a particular diameter rotary file, the required number of teeth shall be not less than the lower number shown and not greater than the larger number shown.

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TABLE III. Number of teeth per circumference, type I rotary files.

Cutter head diameter (inches)	Number of teeth		
	Class a (coarse)	Class b (medium)	Class c (fine)
1/4	10 - 13	14 - 16	17 - 21
1/2	14 - 20	21 - 26	27 - 35
3/4	18 - 24	25 - 32	33 - 44

TABLE IV. Number of teeth per circumference, type II rotary files.

Cutter head diameter (inches)	Number of teeth		
	Class a (coarse)	Class b (medium)	Class c (fine)
3/32	7 - 11	12 - 15	16 - 20
1/8	8 - 11	12 - 15	16 - 20
1/4	12 - 15	16 - 20	21 - 26
3/8	13 - 18	19 - 26	27 - 38
1/2	16 - 22	23 - 28	29 - 39
3/4	19 - 28	29 - 38	39 - 49

3.6 Fluting. The fluting of the rotary file shall be either right hand spiral, right hand spiral with chipbreaker, double cut, or diamond cut depending upon the style specified. The fluting shall be hand ground or machine ground and shall create a cutting tooth pattern that extends along the entire periphery of the cutting head.

3.6.1 Style A fluting. The style A rotary files shall have right hand spiral fluting consisting of a standard commercial right hand helix. The helix angle measured from the centerline of the cutter head shall be not less than 15° and not greater than 35°.

3.6.2 Style B fluting. The style B rotary file shall have right hand spiral fluting with chipbreaker consisting of standard right hand spiral fluting with an additional series of slight serrations along the periphery of the cutting teeth that serve to break up the formation of long chips.

3.6.3 Style C fluting. The style C rotary file shall have double cut fluting consisting of a series of left hand spirals in conjunction with the standard right hand spirals which results in a wedge or chisel form cutting tooth. The number of left hand spirals shall be not less than 40 percent and not greater than 60 percent of the number of right hand spirals.

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3.6.4 Style D fluting. The style D rotary file shall have diamond cut fluting consisting of a series of left hand spirals in conjunction with the standard right hand spirals which results in a pyramidal form cutting tooth that may have either a sharp pointed top or a truncated top. The number of left hand spirals shall be not less than 70 percent and not greater than 90 percent of the number of right hand spirals.

3.7 End fluting. The rotary file shall be furnished with a plain end that shall have no fluting applied to the plane perpendicular to the axis of the tool at the end of the cutting head. When specified (see 6.1), the rotary file shall be furnished with an end cut consisting of a flute pattern applied to the plane perpendicular to the axis of the tool at the end of the cutting head.

3.8 Measurement system. All dimensions in this description are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.9 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practical.

3.10 Workmanship. The workmanship of the rotary files shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in this description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling inspection. The supplier shall randomly select the number of samples in table V that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination in 4.3 and the dimensional inspection in 4.4. Any sample failing to meet all requirements specified herein shall be considered a defective item. If the number of defective items is equal to or less than the quantity specified under the "Acceptance, number defective" column of table V, the supplier shall replace the defective samples with items that meet all the requirements specified herein and the entire lot shall be considered acceptable for shipment. If the number of defective samples is equal to or greater than the quantity specified in the "Rejection, number defective" column of table V, the entire lot shall be considered defective and unsuitable for acceptance. A lot is defined as all rotary files of the same type, style, size, and class presented at one time for acceptance.

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TABLE V. Sampling plan.

Quantity of items being purchased			Sample size	Acceptance, number defective	Rejection, number defective
2	to	8	2	0	1
9	to	15	3	0	1
16	to	25	5	0	1
26	to	50	8	1	2
51	to	90	13	1	2
91	to	150	20	2	3
151	to	280	32	3	4
281	to	500	50	5	6
501	to	1200	80	8	9
1201	to	3200	125	12	13
3201	to	10,000	200	18	19

4.3 Examination. Each rotary file submitted for sample inspection shall be visually examined to determine conformance with all requirements of this description, except for the hardness requirement.

4.4 Dimensional inspection. Each sample rotary file shall be inspected to assure that cutter and shank dimensions are within the tolerances specified herein. Each sample shall be inspected to determine that the number of teeth per circumference conforms to the requirements of table III or IV for the type rotary file offered for inspection. Each sample shall be inspected to determine that the flutes are free of grinding imperfections and conform to the fluting requirements specified herein. The head and shank of each sample rotary file shall be inspected to determine conformance with the concentricity requirements specified herein. To determine conformance to the concentricity requirements, the shadowgraph or dial indicator method shall be used. The shank length of each sample rotary file shall be inspected to determine conformance with the length requirements specified herein.

4.5 Hardness test. After the sample inspection has been completed, a hardness test shall be performed on the nearest whole number to 10 percent of the sample. If any test sample fails to meet the applicable hardness requirements, the total lot from which the sample was selected shall be rejected.

4.6 Packaging inspection. Packaging shall be inspected for conformance with section 5.

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5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or purchase order, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.1), preservation, packing, and marking shall be in accordance with Federal Specification PPP-T-1150, level as specified.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number, and date of this commercial item description.
b. Type, style, size, class, and quantity required (see 1.2).
c. End cut, if required (see 3.7).
d. Packaging in accordance with PPP-T-1150 if required; specify level (see 5.1).

6.2 Cross reference data. The type I, style A, size 5, class a rotary files covered by this description replace the rotary files previously covered by the type VIII, inverted cone rotary files listed in Interim Federal Specification GGG-F-00340. All other types, styles, sizes, and classes covered in this description are new.

MILITARY INTERESTS:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Army - AL
Navy - SH
Air Force - 84
DLA - GS

User Activity

Navy - SH

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3455-0116

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-51166	Files, Rotary, Inverted Cone Shape, High Speed Steel or Tungsten

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)* Carbi
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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